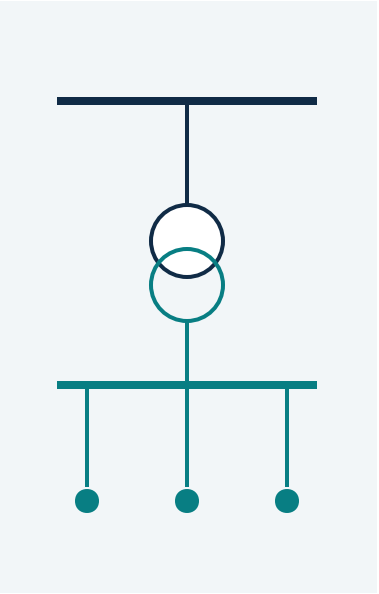
EPG-NSGA-II

Engineering-Prior-Guided NSGA-II for Differentiated Capacity-Load Ratio Optimization in County-Level 110 kV Grids with High Distributed PV Penetration



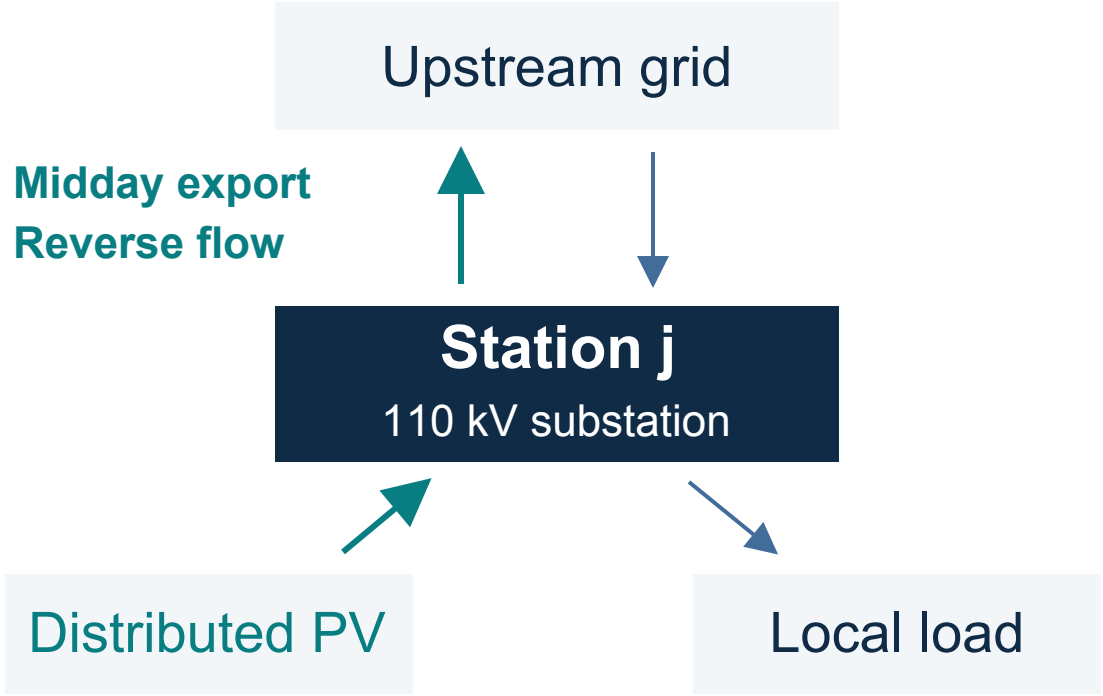
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High PV calls for differentiated planning

A fixed CLR cap of 2.0 misses station-level PV and transfer differences.



JOINT STATION DECISIONS

CLR Rj · Storage Pj (MW)

CLR = planned transformer capacity divided by forecast peak load

MINIMIZE

Annualized cost

million CNY/yr

Expected energy not supplied

EENS · MWh/yr · N−1 inadequacy

Feasibility: station reverse-flow limits · aggregate CLR band 1.80–3.00
Upstream reverse-flow limit included when enabled

Source: camera-ready paper, Introduction and Section II-A. CLR is dimensionless.

Engineering priors guide the search

Standard NSGA-II + **seeding** + **feasibility repair**

Seed & repair
Initial population

Variation
Standard operators

Repair
After variation

Evaluate & select
Cost, EENS, feasibility

Next generation: return to variation

REPAIR PRIORITIES

1 Aggregate CLR

Bounded common offset

2 Station reverse flow

Increase storage power

3 Upstream flow

Allocate storage, if enabled

Remaining violations use constraint domination; standard ranking and selection are retained.

Source: camera-ready paper, Section II-B and Algorithm 1.

Better search, lower modeled planning cost

Search quality | K = 40

+4.8%

vs standard NSGA-II
mean final hypervolume

Method	HV	Gen.
NSGA-II	1.000	800
NSGA-III	0.985	800
Seeding-only	0.998	400
Repair-only	1.048	400
EPG-NSGA-II	**1.048**	**400**

HV normalized to NSGA-II at termination; common reference point.
Population 100 · Three independent runs · Repair nearly matches the full method.

Planning outcome | K = 20

−3.4%

annualized cost

Annualized cost (million CNY/yr)

236.40 → **228.43**

Modeled EENS (MWh/yr)

561.00 → **0.00**

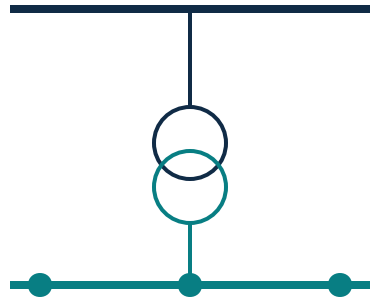
2.0-capped → elastic CLR design

Public-data benchmark · Energy/SOC limits and real-grid calibration remain future work.

Source: camera-ready paper, Table I and Section III-C. Values checked against archived experimental CSVs.

Thank you

Questions & discussion



Engineering priors → **more effective search**

→ differentiated CLR and storage planning